Conditioning a Record on Its Own Future:
Targeted Accordance and Survivorship-Conditioned Path Measures

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Abstract

A future condition can reshape the statistical profile of earlier events without anything in the future acting on the past: conditioning a law over histories on a future event reweights the entire path, and the Doob transform is the standard form of that reweighting. This paper states the attribute-level consequence and keeps it separate from the inference the consequence does not license. The claim in one sentence: if the realized member of a binary partition of the past is drawn under a law conditioned on a future target, then for every $\kappa > 1$ the realized member is, with probability at least $1 - 1/(1+\kappa)$, within a κ-fold exceedance, in conditioned mass, of its complement, a statement indifferent to the target's tense; a productivity reading needs the prior-odds factor the inequality retains, since with $\mathbb{P}(A) = 0.99$, $\mathbb{P}(B \mid A) = 0.01$ and $\mathbb{P}(B \mid A') = 0.02$ the less productive member is realized about 98% of the time under the target-conditioned draw; and whether a given record is drawn under such a law is a separate hypothesis, tested by comparing measures. Four things follow at their stated grades. The theorem, with its one-line proof. Its specialization to survival targets, where the critical regime forbids conditioning on a null event and the right object is the horizon family $\{T > t\}$ with the Q-process as its limit. The between-model layer: the Bayes factor between a survivorship-conditioned and an unconditioned law, which at finite information is the Doob density itself, so that a historical model comparison and a stochastic h-transform are one object; the realized log-evidence fluctuates and the monotone quantity is the relative entropy, whose rate is the information flux. And the limit on certification: the two laws are mutually absolutely continuous on their specified common support on every finite horizon and singular only at infinity, so no event of bounded duration separates them with probability one, while statistical discrimination remains available and is what the Bayes factor measures. Read from the other side, the conditioned mass scores futures: the inhabitability of a candidate future is the weight it gives one's realized record, scoping the future caps one's credence in it at that weight, and across competing futures the inhabitabilities are normalized against prior weights; targeted accordance is the scoping of one's own record pinned at each attribute of its future. A conjunction of predeclared attributes sharpens the theorem multiplicatively under submultiplicativity, and the multiplicity guard that any retrospective scan owes is stated as part of the result. The theorem holds for any fixed target, including one an analyst selected after seeing the record; prospective fixation is a restriction on the application, not a consequence of the theorem. The multiplicity bound counts violations

*Email: akb9408@rit.edu. The survivorship model and its Q-process limit are proved in a companion in applied probability [1] and imported here; the attribute-level apparatus descends from the Nexic companion [2]. The empirical hypothesis that the record of natural history is drawn under the survivorship-conditioned law, and its consequences, are developed in a separate downstream companion on the Route World hypothesis, not cited here so that the dependency runs one way, and the theorems here and the hypothesis there carry their different grades separately. Drafting collaboration: Anthropic Claude (Opus 4.x, Fable 5), under the author's direction.

under the proposed target-conditioned law and is not a false-positive calibration for accepting that law against an unconditioned rival, so the protocol requires predictive probabilities under both competing laws, not baseline priors and a target label alone.

1 Introduction

Conditioning a stochastic process on a future event is standard. The process conditioned to avoid absorption indefinitely, the Q-process, is constructed as a limit of finite-horizon conditionings and realized as a Doob h-transform by the persistence-harmonic function; the penalization literature treats a broad class of such conditionings uniformly [3, 4, 5]. What is standard in probability has a consequence in the interpretation of records that is not usually drawn, and this paper draws it at the level of attributes. If a history is sampled under a law conditioned on a future target, then each realized attribute of that history, read against its complement, is constrained by the target in a way the unconditioned law does not impose: the realized member is unlikely to be greatly outweighed by its complement under the target-conditioned measure, and a quantitative bound says how unlikely; the productivity reading needs the prior-odds factor the bound retains. Nothing in the bound requires the future to act on the past. The conditioning is a statement about which histories are in the support and with what weight, and the past of a weighted history carries the weight's fingerprints.

The paper's discipline is to keep two statements apart that are easy to run together. The first is a theorem: *under* a target-conditioned draw, realized attributes accord with the target. The second is an inference: *because* a realized attribute accords with a target, the record was drawn under conditioning on it. The second does not follow from the first. It is a comparison between rival laws for the record, and it is answered by a Bayes factor, not by the theorem. Both are stated here, in that order, and the theorem's proof is placed beside the disclaimer of the inference so that the two cannot be confused.

The claim, in one sentence. If the realized member of a binary partition of the past is drawn under a law conditioned on a future target, then for every $\kappa > 1$ it is, with probability at least $1 - 1/(1+\kappa)$, within a κ-fold exceedance, in conditioned mass, of its complement, a statement indifferent to the target's tense; a productivity reading needs the prior-odds factor the inequality retains, since with $\mathbb{P}(A) = 0.99$, $\mathbb{P}(B \mid A) = 0.01$ and $\mathbb{P}(B \mid A') = 0.02$ the less productive member is realized about 98% of the time under the target-conditioned draw; whether a given record is drawn under such a law is a separate hypothesis, tested by comparing measures.

What is imported. The survivorship companion [1] proves the model in which survival targets are specialized here, the finite-horizon conditioned laws and their Q-process limit in the critical regime, and the theorem that the limit is locally absolutely continuous with respect to the unconditioned law on survivors and globally singular. The Nexic companion [2] supplies the attribute-level setting: an information state as a set of stated attributes with the unstated blank, the postulate that a realized attribute is drawn by its parent-conditional weights, and the exceedance modality. Each is restated in Section 2 at its grade.

What is new. The theorem in its general form (Section 3), stated for an arbitrary target and proved in one line, with its prior-bound corollary in joint-event form and its conjunction form under the correct dependence assumption. The specialization to survival targets through the horizon family (Section 4), which removes the zero-probability conditioning that a naive statement of the survival target commits. The between-model layer (Section 5): the Bayes

factor between conditioned and unconditioned laws identified with the Doob density at finite information, and the information flux as the monotone statistic the realized evidence lacks. The scoping of a candidate future for inhabitability (Section 6), the theorem's conditioned mass read as a score on futures, with its exact relation to Bayes's theorem and with targeted accordance as its instance at one's own record. The limit on certification (Section 7) stated at its exact strength. And the protocol under which the theorem's application to a real record is a prospective test rather than a retrospective fit (Section 8).

Position in the branch. This paper's core theorem (Sections 2–3) depends on nothing Golden and on no sibling; it is a statement about arbitrary target conditioning. Its survival specialization (Section 4) imports the survivorship companion [1], which is the root of the branch; a downstream technoanatomy companion imports this paper and the survivorship one for its mechanism; and a terminal Route World companion imports all three to state its hypothesis. Neither downstream paper is cited here, so that the citation graph, like the logical one, runs one way. No downstream paper is cited here as support for anything. The scoping of Section 6 depends on nothing Golden either; its examples name the program's uses without consuming them.

What is refused. No claim that any actual record is drawn under a future-conditioned law is made here; that is the downstream companion's hypothesis, and it is carried there at a hypothesis's grade. No retrocausal mechanism is invoked anywhere. And no result below certifies anything about a record from finite data, which Section 7 proves impossible.

2 Setting

Definition 2.1 (Laws over histories, and partitions of the past)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry a law over histories, with a filtration $(\mathcal{F}_t)$ recording the history to time t. An *attribute* is an $\mathcal{F}_\infty$-measurable event; a *binary partition* of the past by an attribute is a pair $\{A, A'\}$ with $A' = \Omega \setminus A$ and $0 < \mathbb{P}(A) < 1$. The *realized member* A^* is the member of the pair that obtains, and $A^{*\prime}$ its complement. A *target* is an attribute B with $\mathbb{P}(B) > 0$, and the *B-conditioned law* is $\mathbb{P}_B := \mathbb{P}(\cdot \mid B)$. Nothing in the definition orders A and B in time: an attribute of the history's future is an attribute.

Definition 2.2 (Information states and the draw; from [2])**.** An information state fixes a set of stated attributes and treats the unstated as blank rather than as unspecified. The *draw* is the postulate that at each open attribute the realized member is distributed by the parent-conditional weights of the law in force; under $\mathbb{P}$ the weights are $\mathbb{P}(A), \mathbb{P}(A')$, and under $\mathbb{P}_B$ they are $\mathbb{P}(A \mid B), \mathbb{P}(A' \mid B)$. The modality $\lesssim$ used below is the *exceedance form*: $X \lesssim Y$ abbreviates the family of statements that $X \geq \kappa Y$ has probability at most $1/(1+\kappa)$ for every $\kappa > 1$, the probability taken over the draw.

Definition 2.3 (The survivorship model; from [1])**.** A history carries a hazard process $h_t > 0$ and a killing time T with $\mathbb{P}(T > t \mid \mathcal{F}^h_\infty) = e^{-\Lambda_t}$, $\Lambda_t = \int_0^t h$. The finite-horizon conditioned laws are $Q_t := \mathbb{P}(\cdot \mid T > t)$. In the *critical regime*, $\mathbb{P}(T = \infty) = 0$ while $\mathbb{E}[T] = \infty$, and the Q-process is the limit $Q := \lim_{t\to\infty} Q_t$ on each $\mathcal{F}_s$. Everything in this paper is written for the abstract density

$$Z_s := \frac{dQ}{d\mathbb{P}}\Big|_{\mathcal{F}_s},$$

and two specializations are imported at their grades: the hard-wall law, proved, $Z_s^{\mathrm{hard}} = \mathbf{1}_{\{\tau > s\}} \mathfrak{h}(X_s)/\mathfrak{h}(X_0)$ with $\mathfrak{h}$ the persistence-harmonic function; and the soft-wall law, conditional

on the companion's ratio hypothesis (its Ansatz B), $Z_s^{\mathrm{soft}} = e^{-\int_0^s q(X_u)du}\, g(X_s)/g(X_0)$, which does not coincide with the hard density even when g is proportional to $\mathfrak{h}$ after a translation: the hard density carries a survival indicator and vanishes on paths crossing the wall, the soft density carries an exponential killing factor and remains positive on finite paths with finite accumulated killing, and the two conditioned laws have different supports. Both densities are placed on the underlying diffusion filtration, the hitting time τ being observable from the path, with the hard hitting indicator in one and the integrated soft-killing factor in the other; a separately observed Cox clock would need its own explicit treatment, and none is given here; the results are established for a specified density Z_s and then specialized twice. No result below depends on which specialization is in force.

Theorem 2.4 (Locally ours, globally elsewhere; **[P]** for the hard-wall law, **[C]** for the soft-wall law, in [1])**.** *In the critical regime $Q|_{\mathcal{F}_s} \ll \mathbb{P}|_{\mathcal{F}_s}$ for every finite s, with mutual absolute continuity on the survival event, while $Q \perp \mathbb{P}$ on $\mathcal{F}_\infty$.*

3 Future-targeted accordance

Theorem 3.1 (Future-targeted accordance)**.** *Let $\{A, A'\}$ be a binary partition of the past and B a target, and suppose the realized member A^* is drawn under $\mathbb{P}_B$. Then for every $\kappa > 1$,*

$$\mathbb{P}_B\left[\frac{\mathbb{P}(B \mid A^{*\prime})}{\mathbb{P}(B \mid A^*)} \;\geq\; \kappa\,\frac{\mathbb{P}(A^*)}{\mathbb{P}(A^{*\prime})}\right] \;\leq\; \frac{1}{1+\kappa}. \tag{1}$$

Equivalently, under the B-conditioned draw, $\mathbb{P}(B \mid A^{*\prime})/\mathbb{P}(B \mid A^*) \lesssim \mathbb{P}(A^*)/\mathbb{P}(A^{*\prime})$.

Proof. By Bayes, for either member C of the pair, $\mathbb{P}(B \mid C')/\mathbb{P}(B \mid C) = \big(\mathbb{P}(C' \mid B)/\mathbb{P}(C \mid B)\big)\cdot\big(\mathbb{P}(C)/\mathbb{P}(C')\big)$, so the event in (1) is $\{\mathbb{P}(A^{*\prime} \mid B)/\mathbb{P}(A^* \mid B) \geq \kappa\}$, that the realized member carries B-conditioned weight $w := \mathbb{P}(A^* \mid B) \leq 1/(1+\kappa)$. For $\kappa > 1$ the threshold is below one half, so at most one member of the pair has conditioned weight below it. Under $\mathbb{P}_B$ that member, if it exists, is realized with probability equal to its conditioned weight, which is at most $1/(1+\kappa)$; if it does not exist the event is empty. □

Remark 3.2 (What the theorem says, and what it does not). The outer probability in (1) is $\mathbb{P}_B$. The theorem says that a record drawn under conditioning on B has realized attributes that typically accord with B: the realized member of each pair is unlikely to carry very small B-conditioned mass relative to its complement, the exceedance being what the draw prices. It does not say that the more B-productive member is typically the one realized, which the counterexample below defeats without an additional prior restriction; and the result is a Bayesian identity and bound applied to historical interpretation, not a new mechanism of historical selection. It does not say that a record whose realized attributes accord with B was drawn under $\mathbb{P}_B$. That is a comparison between $\mathbb{P}_B$ and rival laws for the record, and it is Section 5. The theorem is a within-model statement; the inference is between-model; and the whole discipline of applying this result to an actual record is to keep them apart. Nothing in the statement or the proof distinguishes a target in the past from one in the future: B is an attribute, and the draw is the same draw.

Corollary 3.3 (The prior bound, in joint-event form)**.** *Fix a member A of the pair with $\mathbb{P}(A) = p < \frac{1}{2}$. Under the B-conditioned draw, the joint event that A is realized and is the less B-productive member has probability at most p:*

$$\mathbb{P}_B\big[\, A^* = A \text{ and } \mathbb{P}(B \mid A') \geq \mathbb{P}(B \mid A)\,\big] \;\leq\; p. \tag{2}$$

Proof. $\mathbb{P}(B \mid A') \geq \mathbb{P}(B \mid A)$ holds exactly when $w := \mathbb{P}(A \mid B) \leq p$, by the Bayes identity in the proof above with $\kappa = (1-p)/p$. Under $\mathbb{P}_B$ the member A is realized with probability w, so the joint event has probability $w\,\mathbf{1}\{w \leq p\} \leq p$. □

The corollary bounds the joint event, realized and unproductive, by the prior; it does not bound unproductivity conditional on having observed A, and the slogan that a rare realized attribute establishes its own significance is not licensed by it: if A is independent of B, the joint event with the non-strict comparison has probability p, while conditional on observing A the productivities are equal with certainty. The corollary is a within-model diagnostic, and it is not evidence for the draw.

Proposition 3.4 (Conjunction: the stockpile)**.** *Let $F_1, \ldots, F_n$ be a predeclared family of attributes with priors $p_i = \mathbb{P}(F_i)$, let $F = \bigcap_i F_i$, and suppose the family is* submultiplicative*, $\mathbb{P}(F) \leq \prod_i p_i$, which holds with equality under independence and holds under negative association. Then, under the B-conditioned draw of the partition $\{F, F'\}$,*

$$\mathbb{P}_B\big[\, A^* = F \text{ and } \mathbb{P}(B \mid F') \geq \mathbb{P}(B \mid F)\,\big] \;\leq\; \mathbb{P}(F) \;\leq\; \prod_i p_i : \tag{3}$$

the joint event that the conjunction is realized and is the less B-productive member of its pair has probability at most the product of the priors, which tightens multiplicatively in n while the complement, the union of the F_i', accumulates weight at least $1 - \prod_i p_i$. Positive association gives $\mathbb{P}(F) \geq \prod_i p_i$ and defeats the bound; it is not a sufficient condition.

Proof. Corollary 3.3 applied to the fixed member F of the partition $\{F, F'\}$ gives the first inequality when $\mathbb{P}(F) < \frac{1}{2}$; the second is submultiplicativity. □

Proposition 3.5 (The multiplicity guard: expectation and tail)**.** *If n partitions are interrogated under the B-conditioned draw and N counts those whose realized member exhibits a κ-fold exceedance of (1), then $\mathbb{E}[N] \leq n/(1+\kappa)$; and, with no assumption on the dependence among the partitions,*

$$\Pr[N \geq m] \;\leq\; \frac{n}{m(1+\kappa)}, \qquad \Pr[N \geq 1] \;\leq\; \frac{n}{1+\kappa}, \tag{4}$$

by Markov's inequality and the union bound respectively. Under independence or a stated dependence structure the tail sharpens. Exceeding the expectation is not significance: a declared family is scored against the tail (4), and a family whose size n is not declared cannot be scored at all.

Proof. Linearity of expectation over the n indicators, each of probability at most $1/(1+\kappa)$ by Theorem 3.1; Markov on N; and the union bound on the n events. □

4 Survival targets and the horizon family

The target of most interest is indefinite survival, and in the critical regime it cannot be conditioned on directly: $\{T = \infty\}$ is null, and $\mathbb{P}(\cdot \mid T = \infty)$ is undefined. The right objects are the horizon events.

Theorem 4.1 (Horizon-indexed accordance; the second statement at the grade of the specialization in force)**.** *Let $B_t = \{T > t\}$ and $Q_t = \mathbb{P}(\cdot \mid B_t)$. For every finite t and every binary*

partition $\{A, A'\}$ *of* $\mathcal{F}_s$*,* $s < t$*, if the realized member is drawn under* Q_t *then* (1) *holds with* $B = B_t$ *and* $\mathbb{P}_B = Q_t$*. In the critical regime, as* $t \to \infty$ *the conditioned law* Q_t *converges to the* Q*-process on* $\mathcal{F}_s$*, and for each cell*

$$\frac{\mathbb{P}(T > t \mid A)}{\mathbb{P}(T > t)} \longrightarrow \frac{Q(A)}{\mathbb{P}(A)} = \mathbb{E}_{\mathbb{P}}[Z_s \mid A], \tag{5}$$

so that

$$\frac{\mathbb{P}(T > t \mid A')}{\mathbb{P}(T > t \mid A)} \longrightarrow \frac{Q(A')/\mathbb{P}(A')}{Q(A)/\mathbb{P}(A)}. \tag{6}$$

Proof. The first statement is Theorem 3.1 with the target B_t. For (5), $\mathbb{P}(T > t \mid A)/\mathbb{P}(T > t) = \mathbb{E}_{\mathbb{P}}[\mathbf{1}_A \mathbb{P}(T > t \mid \mathcal{F}_s)/\mathbb{P}(T > t)]/\mathbb{P}(A)$, and by Definition 2.3 $\mathbb{P}(T > t \mid \mathcal{F}_s)/\mathbb{P}(T > t) \to Z_s$ in $L^1(\mathbb{P})$ on the survival event, giving $\mathbb{E}_{\mathbb{P}}[Z_s \mathbf{1}_A]/\mathbb{P}(A) = Q(A)/\mathbb{P}(A)$; (6) is the quotient of two instances. □

Remark 4.2 (Grade propagation in this section)*.* Every statement in this section that consumes the survivorship companion's conditioned law inherits that law's grade: with the hard-wall specialization the statements are at **[P]**; with the soft-wall specialization they are conditional on the companion's ratio hypothesis, Ansatz B, and carry that conditionality wherever they are used. Theorem 3.1 and its corollaries depend on neither and remain unconditional.

Remark 4.3 (The ideal limit is a limit)*.* Every statement about "conditioning on indefinite survival" in this paper and its companions is to be read through Theorem 4.1: a statement at finite horizon, and then a limit. The Q-process is not a conditional probability on a null event, and expressions of the form $\mathbb{P}(G \mid A)$ with G the event of indefinite survival are shorthand for the normalized limit $L(A) := \lim_t \mathbb{P}(B_t \mid A)/\mathbb{P}(B_t) = Q(A)/\mathbb{P}(A)$ where it exists, which is not in general a probability and can exceed one, so it is written $L(A)$ and not $\mathbb{P}(G \mid A)$, the latter inviting probability manipulations that would be invalid; and not for the raw limit $\lim_t \mathbb{P}(B_t \mid A)$, which in the critical regime is zero for every fixed A of positive probability, since $\mathbb{P}(B_t \mid A) \le \mathbb{P}(B_t)/\mathbb{P}(A) \to 0$; the theorem's ratio requires $Q(A) > 0$ wherever it stands in a denominator. In the survivorship model the persistence-harmonic function $\mathfrak{h}$ is what the normalized limit produces, and the attribute-level ratios are ratios of its averages.

Remark 4.4 (Survival through every horizon, stated as content)*.* Each horizon event $\{T > t\}$ has positive probability, and Q is the law that has absorbed every one of them. Read not as a limit of laws but as the content of an information state, Q states the survival attribute at every horizon at once, every other attribute blank: in set notation $\bigcap_t \{T > t\} = \{T = \infty\}$, null in the critical regime, which is why it is stated through the horizon family and never conditioned on as an event. The Golden program calls this content the *Golden Aethus* at its primitive, and its *Golden Axiom* is that this Aethus is a parent of one's own information state: one's state refines it, decomposing with it as a factor, so that the survival attributes are stated in it. The parent form is what lets a conditioning of the universe be stated at all, an attribution being a relation to an information state and never to an absolute ledger; the bare sentence that the universe is conditioned on indefinite survival is, by contrast, ill-defined. Under the axiom the law of one's own record is Q, which is the hypothesis H_Q of Definition 5.1 stated as content. The axiom is an exposed postulate, carried downstream at the grade of an open hypothesis and nowhere asserted or used here; Theorem 7.1 is why no finite record could certify it, and Section 5 is how a record would bear on it.

5 Rival laws for a record: the between-model layer

Definition 5.1 (Hypotheses about the outer measure)**.** For a record with law $\mathbb{P}$ and a target family B_t, let $H_{\mathbb{P}}$ be the hypothesis that the record's realized attributes are drawn under $\mathbb{P}$ (no conditioning), and H_Q the hypothesis that they are drawn under Q (the survivorship-conditioned law, through its horizon family). Other targets, an observer-existence condition among them, define other hypotheses in the same way.

Proposition 5.2 (The Bayes factor is the Doob density)**.** *For a predeclared record statistic $E \in \mathcal{F}_s$, and with both laws conditioned on the common baseline C of Remark 5.3 wherever an observer's record is compared, the Bayes factor between the conditioned and unconditioned laws is*

$$\mathrm{BF}_{Q:\mathbb{P}}(E) = \frac{Q(E \mid C)}{\mathbb{P}(E \mid C)}, \tag{7}$$

which reads $Q(E)/\mathbb{P}(E)$ when no baseline is shared; and on the full information $\mathcal{F}_s$ it is $Z_s^C = \big(\mathbb{P}(C)/Q(C)\big)\, Z_s$ with Z_s the likelihood ratio of Definition 2.3, the hard-wall Doob density or the soft Feynman–Kac density according to the specialization in force, Z_s itself being reserved for the unconditioned comparison, where it is the factor on $\mathcal{F}_s$ without correction. If E is required by the target, $Q(E) \approx 1$, and rare under $\mathbb{P}$, $\mathbb{P}(E) = q$, then $\mathrm{BF}_{Q:\mathbb{P}}(E) \approx 1/q$.

Proof. (7) is the definition of the Bayes factor for a simple event; on $\mathcal{F}_s$ the factor is the Radon–Nikodym density by definition, and the density's form is Definition 2.3. □

Remark 5.3 (The null for a surviving observer is the survivor law, not raw $\mathbb{P}$). Under raw $\mathbb{P}$ there is positive mass on $\{T \le s\}$, where $Z_s = 0$ and $\log Z_s = -\infty$. That is correct as mathematics and wrong as a null for an observer whose record already exists at s: current survival is evidence both hypotheses share, and a comparison that leaves it in credits the conditioned law for the trivial fact that the observation was made. The null for the observed surviving cohort is $\mathbb{P}_s^{\mathrm{surv}} := \mathbb{P}(\,\cdot \mid T > s)$, against which the likelihood ratio depends on the observation space, and the two must not be confused. On the diffusion filtration $\mathcal{F}_s^X$, where the clock is not observed, an independent Cox clock gives $d\mathbb{P}(\,\cdot \mid T > s)/d\mathbb{P}|_{\mathcal{F}_s^X} = e^{-\Lambda_s}/\mathbb{P}(T > s)$, so that

$$\frac{dQ^{\mathrm{soft}}}{d\mathbb{P}_s^{\mathrm{surv}}}\Big|_{\mathcal{F}_s^X} = \mathbb{P}(T > s)\, \frac{g(X_s)}{g(X_0)}, \tag{8}$$

the killing factor cancelling; this is not $\mathbb{P}(T > s)\, Z_s^{\mathrm{soft}}$, which would double-count the killing. On the enlarged observation space that includes the clock, $\{T > s\}$ is observed and the density must be written on that space, where the hard-wall specialization is $\mathbb{P}(T > s)\, \mathfrak{h}(X_s)/\mathfrak{h}(X_0)$, finite on the whole cohort; the downstream Bayes factors must say which space they are written on. More generally every rival law in a comparison is conditioned on the common baseline C the record fixes, observers existing now and survival to the observation time, and the Bayes factor is $\mathbb{P}(E \mid C, H_1)/\mathbb{P}(E \mid C, H_2)$, so that no part of it is earned by evidence both rivals already had to explain.

Remark 5.4 (Why this and not the theorem carries the inference). A low-probability realization under $H_{\mathbb{P}}$ is not a contradiction of $H_{\mathbb{P}}$; it is evidence against it, and (7) says how much. The attribute-level accordance of Theorem 3.1 tells one what to expect of a record if H_Q holds; the Bayes factor tells one whether H_Q holds better than a rival. A record can accord with Q at every interrogated attribute and still be better explained by a rival law that also predicts those

attributes, which is why the comparison must name its rival and predeclare its statistic. The same object, the Doob density, serves both layers: within the model it is the weight by which the conditioning tilts each path, and between models it is the evidence the tilted path carries.

Definition 5.5 (Realized evidence, information, and flux)**.** Let $Z_t = dQ|_{\mathcal{F}_t}/d\mathbb{P}|_{\mathcal{F}_t}$ and $\ell_t = \log Z_t$, the realized log-evidence. Since Z is a $\mathbb{P}$-martingale, ℓ_t is not monotone along a path: new observations can favor Q, then $\mathbb{P}$. The monotone quantity is the *information*

$$I_t := \mathbb{E}_Q[\ell_t] = \mathrm{D}_{\mathrm{KL}}\big(Q|_{\mathcal{F}_t} \,\big\|\, \mathbb{P}|_{\mathcal{F}_t}\big), \tag{9}$$

nondecreasing in t as the filtration grows, with $\mathbb{E}_{\mathbb{P}}[\ell_t] = -\mathrm{D}_{\mathrm{KL}}(\mathbb{P}|_{\mathcal{F}_t}\|Q|_{\mathcal{F}_t}) \le 0$. This monotonicity, which the proposition asserts for fixed underlying laws and an increasing filtration and for those alone, holds there. If the rival is instead the survivor cohort $\mathbb{P}(\cdot \mid T > s)$ at each observation time, the rival moves and the argument does not apply: with $Q(T > s) = 1$, $\mathrm{D}_{\mathrm{KL}}(Q_s \,\|\, \mathbb{P}(\cdot \mid T > s)) = I_s + \log \mathbb{P}(T > s)$, whose second term can offset the growth of the first, so a sequential comparison must either fix its baseline or prove the result for the moving cohort; on a common event $C \in \mathcal{F}_s$ the likelihood ratio is $Z_s^C = (\mathbb{P}(C)/Q(C))\, Z_s$, not Z_s. Where differentiable, $\mathcal{J}_t := dI_t/dt$ is the *information flux*.

Proposition 5.6 (Discriminating rival laws by the law of the evidence)**.** *Under H_Q the realized log-evidence has mean $I_t \ge 0$ and, in the critical regime of the survivorship model, $I_t \to \infty$; under $H_{\mathbb{P}}$ it has mean $-\mathrm{D}_{\mathrm{KL}}(\mathbb{P}_t\|Q_t) \le 0$. The two hypotheses are therefore discriminated by the distribution of ℓ_t, of which the flux is the summary, and not by any monotone accumulation along a single path, which does not exist.*

Proof. The means are (9) and its $\mathbb{P}$-counterpart; divergence under H_Q in the critical regime follows from Theorem 2.4, since mutual singularity on $\mathcal{F}_\infty$ forces the relative entropy to diverge. □

6 Scoping a future for inhabitability

Theorem 3.1 reads the conditioned mass of a record already in hand under a target law. The same mass, read from the other side, scores candidate futures: a future under which one's record would be a fluke can be held only to the extent of the weight it gives that record. This section states that use as a procedure, gives its exact relation to Bayes's theorem, and shows that targeted accordance is its instance at one's own record. Like the theorem, it depends on nothing Golden; its examples name the program's uses without being consumed.

Definition 6.1 (Inhabitability, and scoping; a procedure)**.** Let A^* be the realized member of a binary partition of one's record, with $A^{*\prime}$ its complement, and let X be a *candidate*: an attribute with $\mathbb{P}(X) > 0$, a future one in particular, or any information state one has not been shown to occupy. Suppose X obtains. The *inhabitability* of X, or its admissibility, is the weight X gives one's realized record,

$$\iota(X) \;:=\; \mathbb{P}(A^* \mid X) \;=\; \frac{1}{1+\kappa_X}, \qquad \kappa_X := \frac{\mathbb{P}(A^{*\prime} \mid X)}{\mathbb{P}(A^* \mid X)}, \tag{10}$$

so that κ_X is the exceedance at which X would place the record. To *scope* X is to weight $\neg X$ with the weight of failing the bound on the supposition of X, the weight $1-\iota(X) = \kappa_X/(1+\kappa_X)$

that X gives the record's complement, and X with what remains: one's credence in X is capped at its inhabitability, $\mathrm{cr}(X) \le \iota(X)$. With c one's confidence in the procedure's application and $\pi_0(X)$ the weight X would carry without it, the scoped credence is at most the linear combination $c\,\iota(X) + (1-c)\,\pi_0(X)$.

Proposition 6.2 (What is proved about the scoping; **[P]**)**.** *In the notation of Definition 6.1:*

(i) Super-uniformity. *Under the X-conditioned draw, for every $u < \frac{1}{2}$, $\mathbb{P}_X\big[\iota(X) \le u\big] \le u$: a record whose inhabitability under X is u is a fluke of level u under X's own law.*

(ii) The cap is Bayes where the candidate is not favored. *For a prior weight $\pi_0 = \pi_0(X)$, the posterior odds of X against $\neg X$ given A^* are the prior odds times $\iota(X)/\mathbb{P}(A^* \mid \neg X)$. Where $\neg X$ makes the record typical, $\mathbb{P}(A^* \mid \neg X) \approx 1$, they are the prior odds times $\iota(X)$, and the posterior credence is at most $\iota(X)$ exactly when $\pi_0 \le 1/(2-\iota(X))$, in particular whenever $\pi_0 \le \frac{1}{2}$.*

(iii) Normalization. *Across a partition of competing candidates $X_1, \dots, X_m$ with prior weights $\pi_1, \dots, \pi_m$, the credences are $\pi_j\,\iota(X_j)/\sum_k \pi_k\,\iota(X_k)$. The inhabitabilities are likelihoods, not a distribution over candidates, and need not sum to one.*

Proof. (i) With $B = X$, the event of (1) is that the realized member carries X-conditioned weight at most $1/(1+\kappa)$, which is $\iota(X) \le u$ with $u = 1/(1+\kappa) < \frac{1}{2}$, and Theorem 3.1 bounds its probability by u. (ii) Bayes's theorem; and with $\mathbb{P}(A^* \mid \neg X) = 1$ the posterior is $\pi_0\iota/(\pi_0\iota + 1 - \pi_0)$, which is at most ι if and only if $\pi_0(2-\iota) \le 1$. (iii) Bayes's theorem over the partition, the likelihood of A^* under X_j being $\iota(X_j)$. □

Remark 6.3 (Targeted accordance is the scoping of one's own record). Pin an attribute B of one's future and take one's own record conditioned on the pin: its inhabitability is $\mathbb{P}(A^* \mid B)$, the conditioned weight w of the proof of Theorem 3.1, which the theorem reads as an exceedance with outer probability $\mathbb{P}_B$ and the scoping reads as a cap on the pin. With A^* rare under $\mathbb{P}$, the cap is the *bound form* of targeted accordance: the credence in unfolding into a future containing B is at most the probability that future gives one's present,

$$\mathrm{cr}(B) \;\le\; \mathbb{P}(A^* \mid B), \tag{11}$$

which is Bayes's theorem at the grade of Proposition 6.2(ii) where B is not favored in advance and the complement makes the present typical, and a commitment of the procedure, not a theorem, where it is stated free of priors. Running the pin over the attributes that might occur in one's future scopes the admissibility of one's own record under each of them. Each pin is then predicted with the confidence administered by its non-contradictory weight, its inhabitability, normalized across competing pins as in Proposition 6.2(iii). This is what the Golden program calls retrocausal accordance, in the deflationary sense of a conditioning on a future attribute and nothing further, and it is counted among the highlights of Golden reasoning.

Remark 6.4 (Two scopings the program runs). The worked case is a scoping at the scale of one life: a standing ability that no task in one's future calls on would be a fluke with respect to that future, so the pin of a life without such a task carries low inhabitability. At the scale of a biosphere, the program's *Paradox of Departures* scopes a *departing* future, one in which one's biosphere departs short of the target. In its general form the record's predeclared structure E, rare under $\mathbb{P}$ with $\mathbb{P}(E) = q$ as in Proposition 5.2, is something a departing future takes no

part in producing, so its inhabitability is $\mathbb{P}(E \mid \text{departing}) \le q/\mathbb{P}(\text{departing}) \approx q$, the departing futures holding nearly all of the unconditioned weight. In its external form the departure is made by a constant-rate external hazard within a window of length ℓ opening after a record of length τ, and under every rate λ the configuration has probability

$$e^{-\lambda\tau}\big(1-e^{-\lambda\ell}\big) \;\le\; \frac{\ell}{\tau}\Big(1+\frac{\ell}{\tau}\Big)^{-(1+\tau/\ell)} \;<\; \frac{\ell}{e\,\tau},$$

the maximum falling at $\lambda = \ell^{-1}\log(1+\ell/\tau)$, near $1/\tau$, since the derivative changes sign once there and $(1+1/r)\log(1+r) > 1$ for every $r > 0$: a low rate makes the timing the fluke, a high one the record's length. The display bounds the *joint* configuration, surviving to τ and then departing within ℓ. It does not bound the probability of departing within ℓ conditional on the survival to τ already in hand, which under the rate λ is $1-e^{-\lambda\ell}$, near one for $\lambda \gg 1/\ell$ and small only for $\lambda \lesssim 1/\ell$; the two differ by the factor $e^{-\lambda\tau}$, and the uniformity in λ belongs to the joint weight alone. The step from the joint weight to a cap on the credence in near-term departure is a scoping in the sense of Definition 6.1, the departing future taken as the pin and the record's weight under it as the cap, and it has the grade Remark 6.3 gives every pin: Bayes where the pin is not favored in advance and its complement makes the record typical, a commitment of the procedure where it is stated free of priors. A reader who conditions on the survival first and asks for the hazard in the window is asking a question the display does not answer, and the independent justification of the step is the downstream argument's burden. The program's conclusion from these scopings, that the record caps the weight of departing and so of missing its target, is a hypothesis-level argument made downstream at its own grade; nothing here asserts it.

Remark 6.5 (Scoping inherits the protocol). The present's description A^* is fixed before any pin is read, as the target is fixed before the record is read in Section 8; otherwise a description can be tailored to any pin. And a pin selected because a scan found its inhabitability extreme carries the scan's multiplicity, as an attribute does under Proposition 3.5.

7 The limit on certification

Theorem 7.1 (No bounded observation certifies the law; grade as Theorem 2.4)**.** *In the critical regime, for every finite s, no event in $\mathcal{F}_s$ has probability one under Q and probability zero under $\mathbb{P}$ on the survival event, nor conversely. Consequently no observation of bounded duration certifies whether a surviving record is drawn under Q or under $\mathbb{P}$; certification is available only on $\mathcal{F}_\infty$, where the laws are singular.*

Proof. Mutual absolute continuity on the survival event at every finite s is Theorem 2.4; an event of probability one under one law and zero under the other would violate it. □

Remark 7.2 (Certification against discrimination). The theorem forbids perfect separation and nothing more. Mutually absolutely continuous laws can be statistically far apart, and Proposition 5.2 is exactly the measure of how far: a Bayes factor of 10^8 on a finite window is discrimination of a high order, and it is not certification, because the same window has positive probability under both laws. The theorem is turned on this paper's own subject matter deliberately: whatever a record's attributes say in favor of H_Q, they say it at a Bayes factor and never at probability one.

8 A protocol for applying the theorem to a record

The theorem applies to any record with a law and a target, and a protocol applying it states, for each predeclared attribute, the predictive probability of each outcome under both hypotheses, the Golden target and the observer-conditioned rival, before the observation interval opens; a protocol supplying only one of the two predictive distributions is not a test. Its application to an actual record faces the retrospective-target objection in its full force: any realized history, read backward, contains rare attributes that are prerequisites of whatever it happens to contain, so a set of attributes selected after the fact and found to accord with a target chosen after the fact establishes nothing. Two of the objection's three freedoms are closed by the application's discipline and not by the theorem's form, which holds for any fixed target, including one selected after the record is read: the target must be fixed before the record is read, as a requirement on the application, and the alternative to each attribute is its complement, not an analyst's choice of class. The third freedom, which attributes to interrogate, is not closed by any theorem, and the multiplicity guard prices it only when the scan is declared. The protocol is therefore prospective.

Definition 8.1 (Prospective test of a target hypothesis)**.** Declare, before the observation interval: (i) the target family B_t and the rival law; (ii) the attribute family $F_1, \dots, F_n$, with n fixed; (iii) the baseline priors p_i under the rival law and the dependence assumption on the family (independence, or negative association, or a stated bound on $\mathbb{P}(\bigcap F_i)$); (iv) the record statistic E whose Bayes factor (7) will be reported; (v) the stopping rule and the multiplicity budget $n/(1+\kappa)$ at the κ to be scored. Then observe. A retrospective exercise on an existing record, however suggestive, is hypothesis-generating and is reported as such.

A scoping of futures (Section 6) is run under the same declarations: the description of the present fixed before any pin is read, the pins declared as a family with their prior weights, and each pin's inhabitability reported with the normalization across the family.

9 Open problems

Every open item is gathered here. The soft-wall ratio hypothesis on which the survivorship model's Q-process rests is open in the companion [1], and Theorem 4.1's second statement inherits it. The convergence of Q_t on a window shifted to the present, the law of $(X_{t+u})_{u \le s}$ given survival to t, is a different statement from convergence on an early window and is not established; every claim about a surviving record's *future* behavior waits on it. A finite, monotone evidence budget under which a Copernican position statement would be well posed is not defined, and no Lindy-type result is claimed. And the extension of the multiplicity guard from a fixed family to a sequentially chosen one, with a stopping rule, is stated as a protocol requirement rather than proved as a bound. The scoping's cap stated free of priors is a commitment of the procedure and not a theorem: it is Bayes's theorem where the candidate is not favored in advance (Proposition 6.2), and what licenses it elsewhere is open.

10 Conclusion

The claim stated at the outset stands as proved. If the realized member of a binary partition of the past is drawn under a law conditioned on a future target, then for every $\kappa > 1$ it is,

with probability at least $1 - 1/(1 + \kappa)$, within a κ-fold exceedance, in conditioned mass, of its complement, and the statement is indifferent to the target's tense; a productivity reading needs the prior-odds factor the inequality retains; whether a given record is drawn under such a law is a separate hypothesis, tested by comparing measures, and the measure of that comparison is the Doob density. The future does nothing to the past in any of this. A conditioned law has a support and a weight, and the past of a weighted history carries the weight; that is the whole of what "conditioning a record on its own future" means, and it is enough to make the companion's hypothesis about the record of natural history a hypothesis with a stated test rather than a reading with a stated feeling. The same weight, read from the other side, scores futures: a future under which one's record would be a fluke can be held only to the extent of the weight it gives that record, and targeted accordance is that scoping run over one's own future, pin by pin.

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